# UMC 0.11um LOGIC/MIXED_MODE Al Advanced Enhancement High Speed 1.2V MOSFET SPICE Model With Thermal Noise Enhancement Document.

**Version 1.0 Phase 4**
**2015/04/08**

# Table of Contents

# 1 Introduction

**Table 1-1 Revision history**

| Version | Date | Author | Remark |
|---|---|---|---|
| 0.5_p1 | 2010/12/24 | YZ Wang | 1. Added thermal noise enhancement model.<br>2. Revised the target values (10um/0.116um) based on EDR revision.<br>3. Triple well NMOS model was copied from twin well NMOS model. |
| 1.0_p1 | 2012/09/14 | Laney Fan | 1. Version up to 1.0 based on 6 pcs Si data. |
| 1.0_p2 | 2013/04/15 | YZ Wang | 1. Modify the Verilog-A file. |
| 1.0_p3 | 2015/02/05 | Kelly Pao | 1. Modify subckt naming of tn model. |
| 1.0_p4 | 2015/04/08 | Kelly Pao | 1. Modify subckt naming of igate model. |

## 1.1 General information

This document describes MOSFET model with thermal noise model enhancement for UMC 0.11 um Logic and Mixed-Mode 1P8M Metal Metal Capacitor Al Advanced Enhancement Process. At the current release, Spectre, Hspice and Eldo models are supported.

Usually BSIM3 built-in thermal model can model the channel thermal noise reasonably well for longer channel devices, however it is difficult to predict the channel thermal noise for shorter channel device over different dimensions and biasing conditions.

In this thermal noise model enhancement, we use a propriety method to compensate the excess channel thermal noise of short channel devices for BSIM3 built-in thermal noise model. The simulated channel thermal noise is compared to the thermal noise extracted from high frequency noise measurement for different device sizes and basing conditions, to make sure the model scalabilities for device dimension and biasing conditions.

Except noise behavior, the models with and without thermal noise enhancement have exact same electrical behaviors.

However, the simulation using the model with the thermal noise enhancement generally needs more simulation time than the simulation using the original model without THERMAL noise enhancement, since Verilog-A code is used in the thermal noise enhancement model. It is recommended to use the model without thermal noise enhancement for simulations other than noise analysis, and use the model with thermal noise enhancement for noise analysis to get more accurate noise results.

Please note that the models with thermal noise enhancement need Verilog-A licenses from the simulators.

## 1.2 Model Usage

### 1.2.1 Model Files

The following model files are included in the model packages,

**Simulator Versions:**

Synopsys HSPICE release 2007.09
Cadence SPECTRE version 6.2.1.299
Mentor Graphics ELDO version 6.11_1.1

**Note:** Cadence claimed ( on 2010/12/06 ) that a bug of thermal noise model in Spectre will be fixed in the upcoming release of MMSIM72_ISR16.

**HSPICE**

l110ae_hs12_v104.lib - 5-libs w/o tn enhancement, 5-libs with tn enhancement
l110ae_hs12_v104.mdl - model parameter file w/o tn enhancement
l110ae_hs12_v104_tn.mdl - model parameter file with tn enhancement
l110ae_hs12_v104_h.va - Verilog-A code

**SPECTRE**

l110ae_hs12_v104.lib.scs - 5-libs w/o tn enhancement, 5-libs with tn enhancement
l110ae_hs12_v104.mdl.scs - model parameter file w/o TN enhancement
l110ae_hs12_v104_tn.mdl.scs - model parameter file with tn enhancement
l110ae_hs12_v104_s.va - Verilog-A code

**ELDO**

l110ae_hs12_v104.lib.eldo -5-libs w/o tn enhancement, 5-libs with tn enhancement
l110ae_hs12_v104.mdl.eldo - model parameter file w/o tn enhancement
l110ae_hs12_v104_tn.mdl.eldo - model parameter file with tn enhancement
l110ae_hs12_v104_e.va - Verilog-A code

As below, 5 libraries without thermal noise enhancement are tt, ss, ff, snfp, fnsp, and 5 libraries with thermal noise enhancement are tt_tn, ss_tn, ff_tn, snfp_tn, fnsp_tn,

tt: typical n-ch mosfet and typical p-ch mosfet
ff: fast n-ch mosfet and fast p-ch mosfet
ss: slow n-ch mosfet and slow p-ch mosfet
fnsp: fast n-ch mosfet and slow p-ch mosfet
snfp: slow n-ch mosfet and fast p-ch mosfet
tt_tn: typical n-ch mosfet and typical p-ch mosfet with thermal noise enhancement
ff_tn: fast n-ch mosfet and fast p-ch mosfet with thermal noise enhancement
ss_tn: slow n-ch mosfet and slow p-ch mosfet with thermal noise enhancement
fnsp_tn: fast n-ch mosfet and slow p-ch mosfet with thermal noise enhancement
snfp_tn: slow n-ch mosfet and fast p-ch mosfet with thermal noise enhancement

### 1.2.2 The definitions of sub-circuit model parameters

After invoking thermal noise enhancement, the compact transistor models evolve to sub-circuit models. The input parameters for the sub-circuit model are the same as the compact model, except the input parameter "mf" for Hspice and Eldo.

The following is the parameter list:

- nf: number of gate fingers
- wf: gate width
- lf: gate length
- as: source area per finger
- ad: drain area per finger
- ps: source periphery per finger
- pd: drain periphery per finger
- sa: distance between OD edge to poly from one side
- sb: distance between OD edge to poly from other side
- sd: poly space for multiple finger device
- mf: multiple factor used for thermal noise simulator (for HSPICE and Eldo only, not required by Spectre). Its value should be always equal to instant multiple factor parameter "m"

### 1.2.3 Multiple Factor "mf" for Hspice and Eldo

Due to the fact that Verilog-A codes in Hspice and Eldo don't support multiple factor that is inherited from the netlist, an additional model input parameter "*mf*" has to be defined in order to accurately simulate the High Frequency noise for multiple devices connected in parallel. *mf* will not affect the simulation results other than thermal noise behavior. The value of *mf* should be always be equal to the value of instant multiple factor parameter *m*. Please refer to the following example:

If 4 devices are connected in parallel, the netlist for Hspice and Eldo should be:

*x1 (d g s b) n_12_hsl110e_ig  l=0.12u w=9.6u nf=4 m=4 mf=4*

# 2 Simulation Results for Channel Thermal Noise

In this section, the simulation results for channel thermal noise against measured channel thermal noise extracted from high frequency noise figure measurement for MOSFET are demonstrated.

o Channel thermal noise extracted from measurement

— Simulated channel thermal noise with thermal noise enhancement

······· Simulated channel thermal noise w/o thermal noise enhancement

Please note in the following figures:

lf=l is the gate finger length

wf=w is the gate finger width

## 2.1 1.2V HS NMOS (*n_12_hsl110e_ig*)

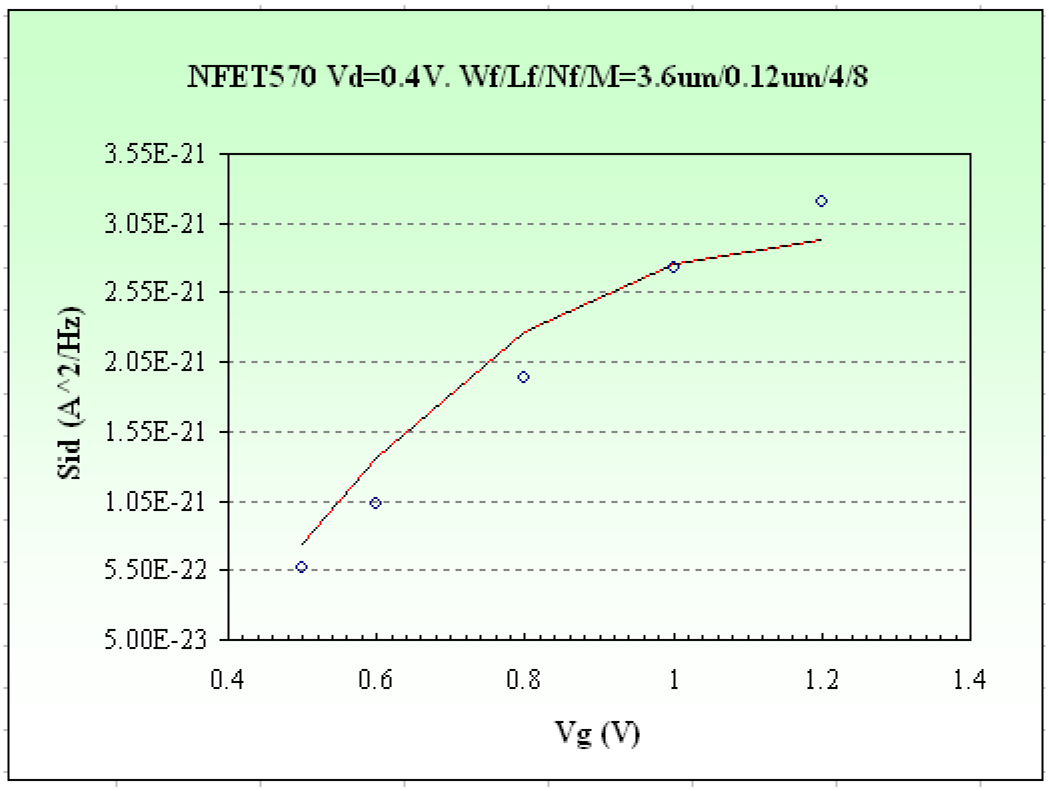


**Figure 2-1a NFET570: Vd=0.4V**

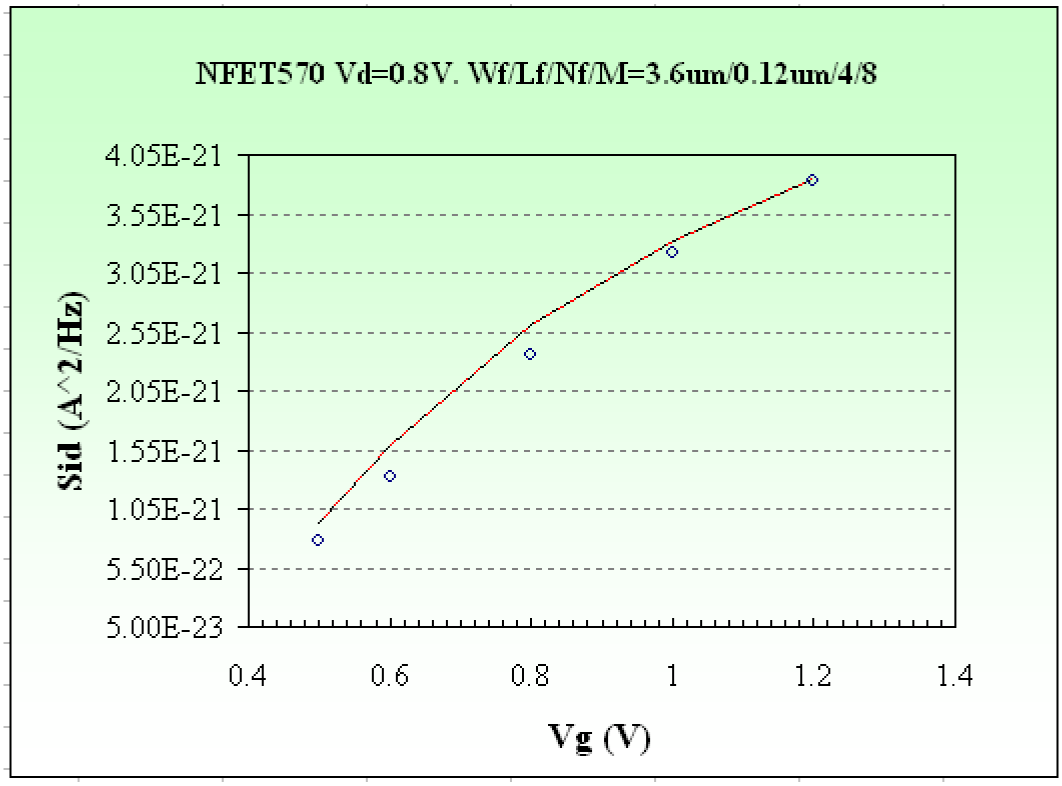


**Figure 2-1b NFET570: Vd=0.8V**

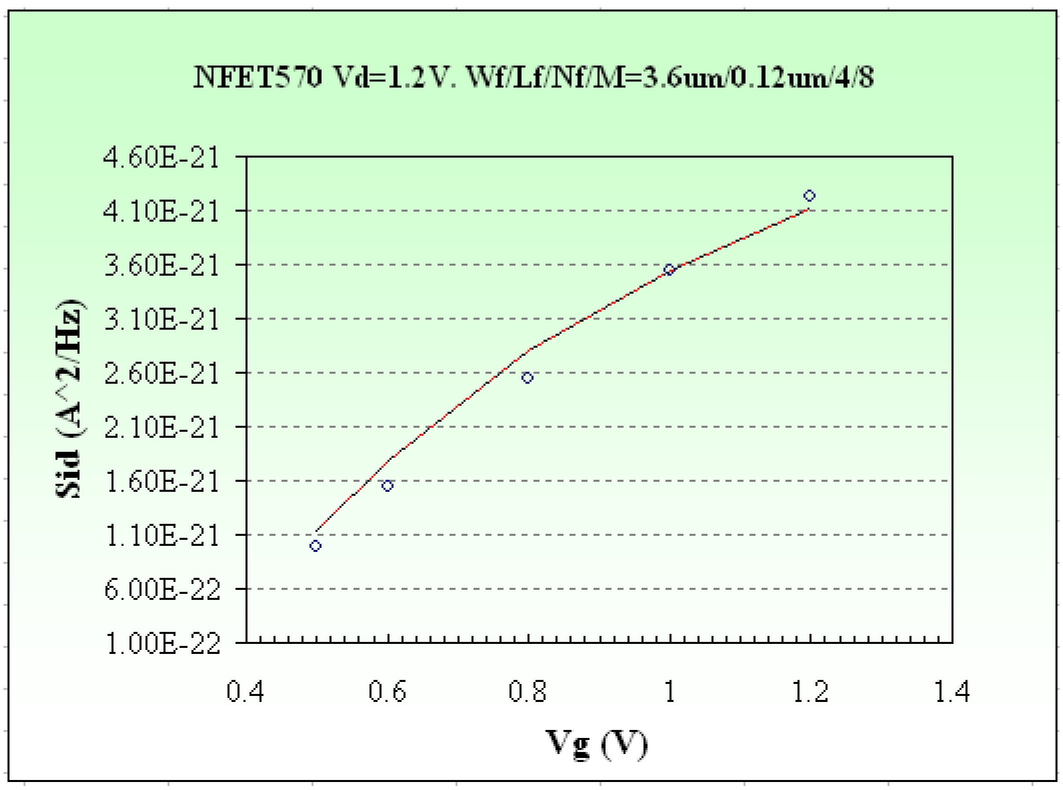


**Figure 2-1c NFET570: Vd=1.2V**

## 2.2 1.2V HS PMOS (*p_12_hsl110e_ig)*

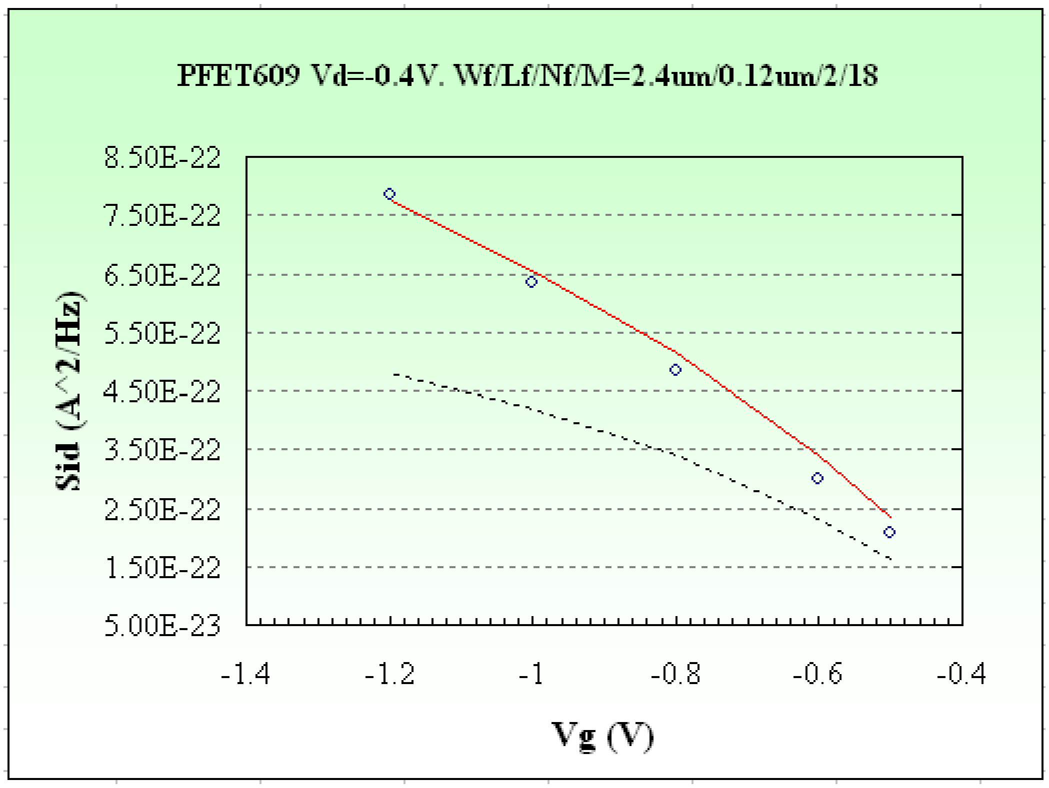


**Figure 2-2a PFET609: Vd=-0.4V**

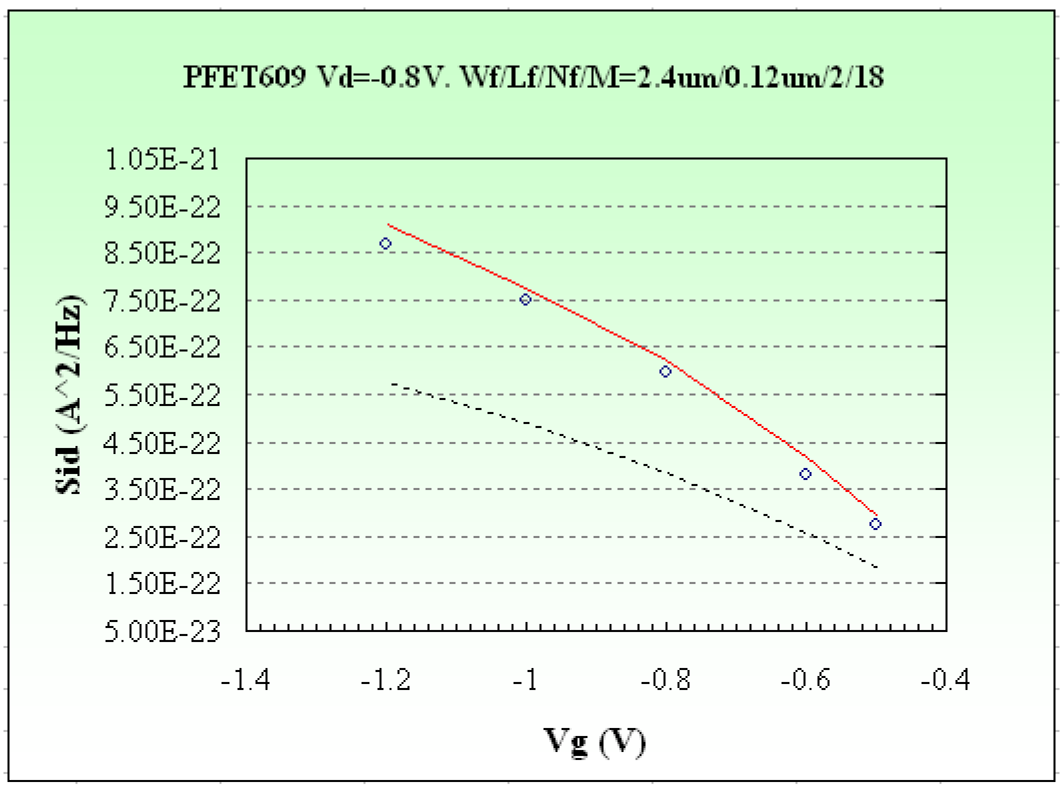


**Figure 2-2b PFET609: Vd=-0.8V**

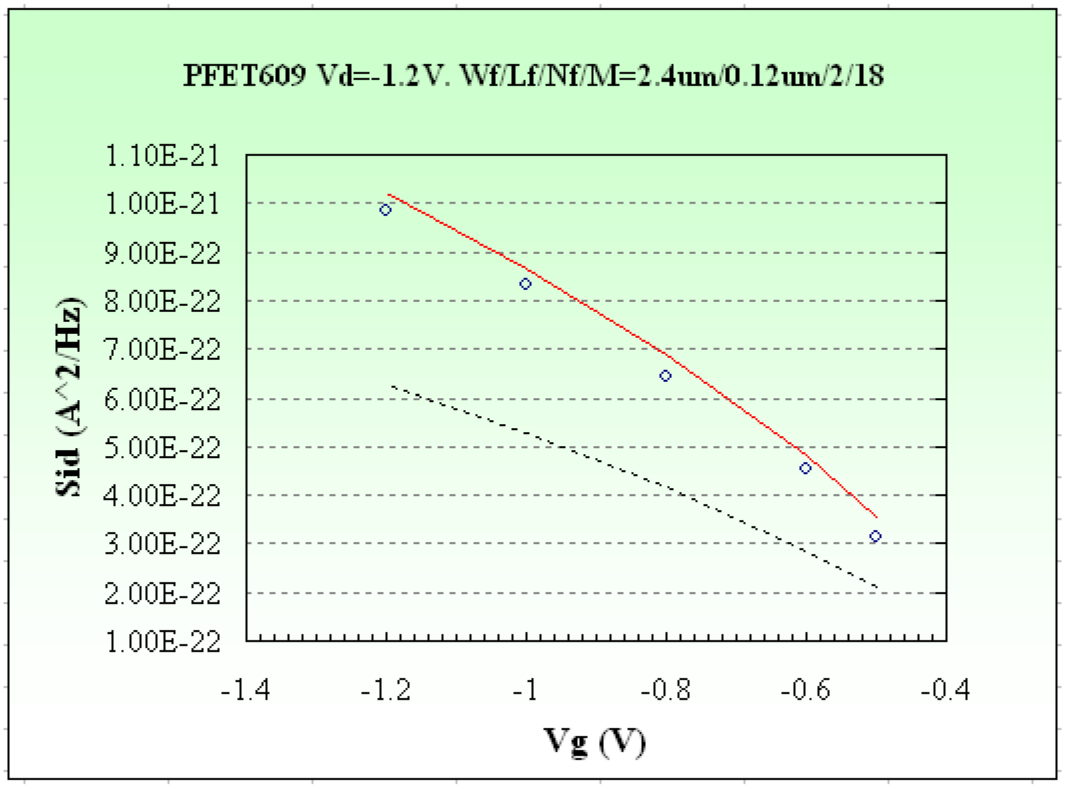


**Figure 2-2c PFET609: Vd=-1.2V**

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